

# Pooja Yogesh Athawale

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## SUMMARY

Data Science and AI Professional with experience delivering data-driven solutions across Healthcare, Analytics, and Cloud Technologies. Proven ability to transform complex data into actionable insights and measurable business outcomes through advanced analytics and intelligent systems.

## TECHNICAL SKILLS

**Programming & Databases:** Python (Pandas, NumPy), SQL, R, PostgreSQL, SQLite

**Data Analytics & Visualization:** Statistical Analysis, Feature Engineering, EDA, Data Cleaning, Data Preprocessing, KPI Reporting, Tableau, Power BI, Excel, Matplotlib, Seaborn

**Machine Learning & AI:** Scikit-learn, PyTorch, Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, NLP, Computer Vision, Hugging Face Transformers, LangChain, Sentence Transformers, ChromaDB

**Cloud & Data Engineering:** AWS, Databricks, PySpark, ETL Pipelines, REST APIs, Git, GitHub Copilot, Jupyter Notebook

## AWARDS & PUBLICATIONS

**Honorable Mention for Outstanding Research Presentation**, Arkansas Academy of Science (AAS), 2026

**Published 2 Research Papers**, IntelliSys 2026, Amsterdam, Netherlands

- KL Prior on Latent Space Structure in Breast Ultrasound Representation Learning.
- Robust Diabetic Retinopathy Grading via Energy-Based Margin Learning

## EXPERIENCE

**Research Assistant, Southern Arkansas University, Arkansas, USA**

May 2025 – Jul 2026

- Evaluated Machine Learning, Generative AI, and RAG models on biomedical datasets using Python, PyTorch, and Scikit-learn, generating insights through statistical analysis, feature engineering, and model validation.
- Improved research reproducibility by standardizing data preprocessing, model evaluation, and experimental workflows, contributing to AI research publications.

**Data Analyst, Hiten Technologies, Pune, India**

Oct 2019 – Nov 2021

- Improved reporting efficiency by 40% and data quality by 30% by automating ETL pipelines using Python, SQL, AWS, and data integration frameworks.
- Enabled executive decision-making through Tableau and Power BI dashboards while reducing cloud infrastructure costs by 25% and maintaining 99.9% system availability using AWS-based analytics solutions.

**Senior Executive, iProcess Services Pvt. Ltd., India**

Jun 2019 – Oct 2019

- Supported banking operations by managing customer accounts, transaction records, and regulatory documentation with high accuracy and compliance.
- Improved service efficiency by resolving customer inquiries, maintaining financial records, and supporting daily branch performance activities.

## PROJECTS

**Clinical RAG Reliability Analysis** – Built a Clinical RAG pipeline using PubMedQA, ChromaDB, LangChain, Sentence Transformers, and LLMs for healthcare question answering. Evaluated retrieval quality, hallucination detection, and grounding to improve AI reliability.

**Breast Ultrasound Representation Learning** – Developed a PyTorch-based Variational Autoencoder (VAE) on 780+ breast ultrasound images to analyze latent-space representations. Evaluated feature quality using UMAP, t-SNE, clustering metrics, and downstream classification tasks.

**Robust Diabetic Retinopathy Grading via Energy-Based Learning** – Built a ResNet-18 and Energy-Based Learning framework to improve confidence calibration and prediction reliability in retinal disease screening. Applied Energy Gap analysis to identify uncertain cases for human-in-the-loop decision support.

## EDUCATION

**Master of Science in Data Science**, Southern Arkansas University, USA

Jan 2025 – Jul 2026

**Post Graduate Program in Cloud Computing**, The University of Texas at Austin, USA

**Bachelor of Engineering in Information Technology**, Dr. Babasaheb Ambedkar Marathwada University, India

## CERTIFICATIONS

Google AI Professional Certificate | AWS Machine Learning Foundation | AWS Cloud Practitioner | Salesforce AI Associate  
Tableau Desktop Specialist | Scrum Fundamentals | Six Sigma Yellow Belt | Jira & Confluence